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Key Points:

- The release of waste from a decommissioned fertilizer plant significantly increased nitrate uptake in the Kansas River
- Despite high N loading and low water temperatures, the Kansas River removed nitrate at rates equal to or greater than small streams
- NO_3^- uptake was high when metabolic activity was low, suggesting NO_3^- removal was not due to assimilatory activity alone

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

A. J. Burgin,
burginam@ku.edu

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High Supply, High Demand: A Fertilizer Waste Release Impacts Nitrate Uptake and Metabolism in a Large River

Michelle C. Kelly¹ , Lydia H. Zeglin² , Admin Husic³ , and Amy J. Burgin⁴ 

¹Department of Ecology and Evolutionary Biology and Kansas Biological Survey, University of Kansas, Lawrence, KS, USA, ²Division of Biology, Kansas State University, Manhattan, KS, USA, ³Department of Civil, Environmental and Architectural Engineering, University of Kansas, Lawrence, KS, USA, ⁴Departments of Environmental Studies and Ecology and Evolutionary Biology & Kansas Biological Survey, University of Kansas, Lawrence, KS, USA

Abstract Current understanding of the relationship between nitrate (NO_3^-) uptake and energy cycling in lotic environments comes from studies conducted in low-nutrient ($\text{NO}_3^- < 1 \text{ mg-N L}^{-1}$), small (discharge $< 1 \text{ m}^3 \text{ s}^{-1}$) systems. Recent advances in sensor technology have allowed for continuous estimates of whole-river NO_3^- uptake, allowing us to address how the relationship between nutrient uptake and metabolism changes over time and space in larger rivers. We used a six-month, controlled nitrogen (N) waste release into the eighth order Kansas River (USA) as an ecosystem level nutrient addition experiment. We deployed four NO_3^- and dissolved oxygen sensors along a 33 km study reach, from February to May 2018, to assess the spatiotemporal relationship between nutrient uptake and stream metabolism during the waste addition. Contrary to our prediction, we did not find evidence of uptake saturation despite an extreme increase in nutrient supply during winter, a period of generally lower biological activity. Although high uptake rates were observed across the study reach, they were uncorrelated to gross primary production. Overall, despite winter temperatures, NO_3^- uptake rates were high compared to small streams and rivers. We provide evidence that large rivers can be effective ecosystems for retaining and transforming nutrients, while showing that the fine-scale mechanisms that regulate nutrient retention in large rivers are still largely unknown.

Plain Language Summary Understanding the link between nutrient and energy cycling in streams and rivers is crucial for water quality management. Despite calls to action over the last decade, our understanding of these processes in large rivers has lagged behind our understanding in small streams. To discover how a large river may respond to increased nutrient loading, we deployed water quality sensors to study a sustained (~6 months) release of highly nitrogen-concentrated wastewater from a fertilizer manufacturing plant into the Kansas River. We found that organisms nearest the release site took up nitrogen (as nitrate) at the fastest rates, despite potentially saturating levels of nitrogen supply. We found that photosynthesizing organisms (i.e., primary producers, modeled as gross primary production) were likely responsible for most nitrate uptake, but were limited by wintertime cold water temperatures. After the release ended, and water temperatures increased, there was a more positive relationship between primary producer activity and nitrate uptake at the two sites nearest the release point. Nitrate uptake rates were high in comparison to other smaller streams and rivers, despite cold water temperatures which can slow organism growth. Our work shows that large rivers have the ability to retain and remove nutrients at rates similar to or larger than smaller rivers and streams, but that we don't yet understand the fine-scale factors that control nutrient retention in large rivers.

1. Introduction

Synthetic nitrogen (N) fixation for the production of agricultural fertilizer has doubled N supply to aquatic ecosystems (Levin et al., 2009; Meybeck, 1982). From 1908 to 2008, the use of synthetic N fertilizer in agriculture accounted for 48% of world population growth (Erisman et al., 2008). Despite this growth, only 17% of global N fertilizer application (100 Tg-N) is consumed in the production of crops and livestock (Erisman et al., 2008). A large fraction is flushed from the landscape with precipitation, ending up in streams and rivers (Carpenter et al., 1998). Almost 60% of the total annual N delivered from the Mississippi River to the Gulf of Mexico originates from cultivated row-crops, even though this farmland accounts for just 30% of the Mississippi River watershed area (Alexander et al., 2008). This disproportionate loading occurs as highly soluble nitrate (NO_3^-), which readily leaches from agricultural soils into waterways (Elrashidi et al., 2005) where it

is either removed from the ecosystem through denitrification (Mulholland et al., 2008), or incorporated into plant, algal, or microbial biomass (Ensign & Doyle, 2006). Thus, the ultimate effect of increased N supply is eutrophication, acidification, and deteriorating water quality (Smith et al., 1999; Vitousek et al., 1997). In extreme cases, high N export from rivers can lead to hypoxic “dead zones” along coastlines (Rabalais et al., 2002; Scavia et al., 2003).

Despite water quality impairment due to N loading, streams and rivers act as zones of biogeochemical N removal and remediate downstream impact of excess N. Mechanisms of N removal include permanent (e.g., denitrification) and temporary (e.g., assimilatory uptake) pathways. In small streams, approximately 15% of N removal is due to denitrification (Mulholland et al., 2008). The remaining N removal can occur through assimilatory uptake within the channel (Hall et al., 2009) and hyporheic zone (Zarnetske et al., 2011), and longer-term storage in benthos (Bernhardt et al., 2005). Given its control on N removal, understanding what drives assimilatory uptake is crucial to understanding aquatic N cycling. Assimilatory uptake is performed by both autotrophic (e.g., macrophytes and algae) and heterotrophic (e.g., bacteria and fungi) organisms, as they take up bioavailable N from the water column and assimilate it into biomass. Assimilation temporarily lowers in-stream N concentrations, until biota senesce (days to weeks later) and N is re-released (Hefting et al., 2005). We can conceptualize the amount of dissolved inorganic N that is transformed into particulate organic N as the areal uptake rate (U) per unit area per time (Payn et al., 2005). Since U can be strongly affected by ambient nutrient concentrations, we can additionally use uptake velocity (V_p), a measure of biological demand relative to in-stream concentration. Greater uptake velocity signifies quicker nutrient uptake and more efficient nutrient removal (Reisinger et al., 2015; Stream Solute Workshop, 1990; Webster et al., 2003).

Recently, there has been an increased interest in connecting our understanding of lotic energy cycling (e.g., whole-stream metabolism) with novel methods of estimating N uptake from high-frequency sensor data (Bernhardt et al., 2018; Heffernan & Cohen, 2010). Gross primary production (GPP) is one such measure of whole-stream metabolism and reflects photosynthetic activity, as estimated by the diel production of in-stream oxygen (Covino et al., 2018; Heffernan & Cohen, 2010; Lupon et al., 2016; Reijo et al., 2018). As primary producers simultaneously photosynthesize and incorporate available N into biomass, NO_3^- uptake (U) should linearly increase with GPP. Additionally, heterotrophic organisms have a similar capacity to incorporate available N into biomass, which can be reflected in measures of ecosystem respiration (ER). ER is a modeled measure of oxygen consumed by heterotrophs and autotrophs during a day, where greater oxygen consumption can signify greater heterotrophic or autotrophic activity (Christensen et al., 1990). GPP often positively correlates (hereafter, “couples”) with day-night (diel) fluctuations in NO_3^- , suggesting that primary production controls uptake in many streams (Hall & Tank, 2003; Heffernan & Cohen, 2010; Lupon et al., 2016). However, many of these studies have been conducted under conditions that promote high autotrophic growth (e.g., stable hydrology, summertime temperatures, and prolonged daylight) and uptake (e.g., low nutrient concentrations with high biotic demand). There is also a deficit of studies conducted in larger systems, which present logistical challenges for sampling and instrumentation (Hensley et al., 2014; Tank et al., 2008), and are more likely to have anthropogenically altered hydrology (Hall et al., 2015). Given this, our current understanding of the relationship between riverine energy and nutrient cycling is likely biased by the lack of measurements from large rivers.

In this study, we address the knowledge disconnect between small-stream and large-river biogeochemistry by asking: How does the relationship between nutrient uptake and metabolism change over time and space during a nutrient addition in a large river? We examine three hypotheses: (a) Nearest the release point, nitrate uptake will be lowest, due to nutrient saturation, as biota experience a diminished capacity for nutrient uptake. (b) Nearest the release point, where nutrient supply is abundant, gross primary production will be decoupled from nitrate uptake, as nitrate supply outpaces the quantity of nitrate required for GPP. (c) After the nutrient addition ends and nutrient supply decreases, nitrate uptake will re-couple with gross primary production, as biota no longer experience nutrient saturation conditions and nitrate availability becomes more limiting for GPP. To evaluate these hypotheses, we measured diel NO_3^- concentrations together with stream metabolism during and immediately after a six-month N release event on the eighth-order Kansas River.

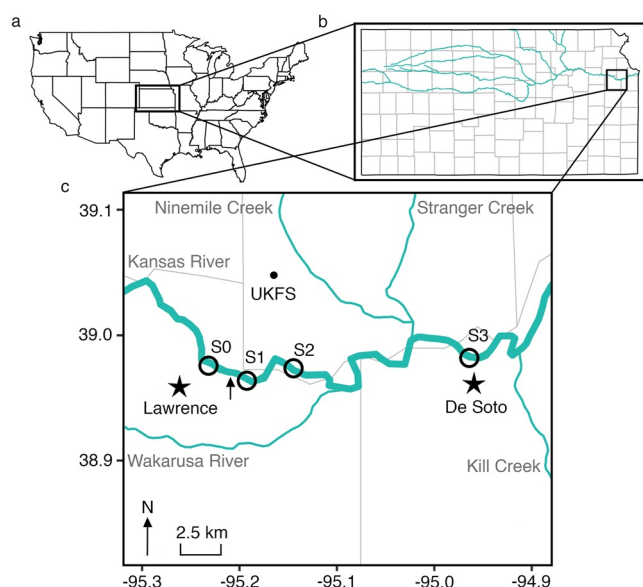


Figure 1. Map of USA (a), the state of Kansas (b), and study reach (c). Blue lines in (b) denote the Kansas River and major tributaries. Blue lines in (c) denote Kansas River and associated tributaries within the study area. Open circles in (c) denote the location of study sites (labeled as S0, S1, S2, S3), “↑” marks the nutrient release site, “Lawrence” marks the city of Lawrence, “De Soto” marks the city of De Soto, “UKFS” marks the location of the University of Kansas Field Station, site of the NEON flux tower from which PAR measurements were obtained. Site S0 located 2.5 km upstream of release outfall, S1 located 0.3 km downstream of outfall, S2 located 5.5 km downstream of outfall, and S3 located 31 km downstream of outfall.

2. Materials and Methods

2.1. Site Description

The Kansas River is a wide (reach average 0.2 km), shallow, prairie-fed river, with uniformly distributed sandy sediments. About 80% of the river's total drainage area (155,000 km²) is captured and controlled by 18 large federal reservoirs, which are released intermittently to maintain navigation on the Missouri River (US Army Corps of Engineers, 1984). The mainstem of the Kansas River is also dammed and hosts the only hydroelectric run-of-the-river dam in Kansas, the Bowersock Dam (built in 1874), in Lawrence. For most of its 272 km length, including the study areas described herein, it is an eighth-order river.

In 2010, the City of Lawrence, KS (“the City”) acquired the property of a former nitrogen fertilizer plant (The Farmland Manufacturing Plant) near the bank of the Kansas River. The fertilizer manufacturing process produced heavily polluted wastewater, which was stored in surface ponds or subsurface tanks. Wastewater ponds and tanks had mean N concentrations ranging from elevated to extremely high (0.15–33,310 mg NO₃[−]-N L^{−1}, 0.06–51,640 mg NH₄⁺-N L^{−1}, as reported by the City), and were on average about twice as DOC rich as the river (12.9 mg L^{−1} vs. 6.4 mg L^{−1}). Previous remediation strategies were not effective at reducing the volume of contaminated water and, in late 2017, near-capacity surface water storage ponds were in danger of uncontrolled overflows. The City requested a one-time controlled release of contaminated water from the facility to the Kansas River, which was approved by the Kansas Department of Health and Environment in September 2017. Over a six-month period (November 2017–April 2018), 114 × 10⁶ L of contaminant were permitted for release from the facility at a rate not exceeding 1.9 × 10⁶ L per day. This release effectively created a whole-ecosystem experiment wherein a concentrated source of N was released to the river, at a relatively constant rate, over a six-month period.

2.2. Sampling Methods

To measure nutrient uptake and stream metabolism, we deployed sensor arrays at three sites (Figure 1; S0, S1, S2) and obtained data from existing USGS stations (S0 and S3). The first site (S0) was located 2.5 km upstream of the waste release and the following three sites (S1, S2, and S3) were located at 0.3 km, 5.5 km, and 31 km downstream of the release, respectively. NO₃[−], DO, and water temperature were measured at all sites and gauge height and discharge (Q , m³s^{−1}) were additionally measured at S0 and S3. These sensor arrays allowed us to evaluate spatiotemporal changes in nutrient supply, uptake, and GPP within the river reach.

Sensor arrays were deployed in the Kansas River (Figure 1) at the intake pool of the Bowersock Dam (S0), East 1625 Rd in Lawrence, KS (S1), and on property located near 246th St east of Fall Leaf, KS (S2) from early February to early June of 2018. Sampling site selection was highly constrained by river access, as the majority of land adjacent to the riverbank is privately owned. River width varied along the study reach and was estimated from USGS/NASA Landsat satellite imagery (S1 = 210.9 m, S2 = 113.4 m, S3 = 138.1 m). Sensors were secured at a distance of about 1 m from the river bank at a depth of about 0.5 m (S1, S2) or against the concrete wall of the dam intake pool at a depth of 1 m (S0). River discharge was historically low during the study period (56.1 m³s^{−1} at De Soto, KS compared to 1917–2018 annual mean of 194 m³s^{−1}) which, when combined with the river's wide channel, made sensor deployment in the advective zone uncertain at sites S1 and S2. Nitrate was measured every 15 min by HACH Nitratax Plus SC (Loveland, CO) UV sensors with an optical path length of 2 mm and a detection limit of 0.1 mg-N L^{−1}, which self-cleaned with a wiper blade every 5 min. NO₃[−] sensors at S0, S1, and S2 were cross-calibrated with the USGS NO₃[−] sensor at S3, maintained by the Kansas Water Science Center in Lawrence, before deployment. Water temperature and DO measurements were recorded every 15 min using a PME miniDOT with an anti-fouling copper plate attachment (Vista, CA).

at sites S1 and S2. A PME miniDOT was not installed at S0 (Bowersock Dam) as DO measurements would be distorted by mixing at the dam intake pool. In-field checks of sensor arrays (e.g., routine cleaning and battery replacement) occurred on a 1.5–2 week period. A water quality sensor array operated by the USGS gaging station at De Soto (S3) recorded dissolved oxygen (DO), NO_3^- , and water temperature measurements at a 15-min temporal resolution.

Measurements of discharge (Q , m^3s^{-1}) at the upstream (Lawrence, KS; S0) and downstream (De Soto, KS; S3) ends of the 33 km study reach were obtained from USGS gaging stations (nos. 06891080, 06892350). Q at site S1 and S2 was estimated by adding the daily volume of waste effluent pumped into the river (as measured by the City of Lawrence) to the USGS gauge reading at site S0. We calculated nitrate load (kg-N day^{-1}) at each site as nitrate concentration times Q . We obtained photosynthetically active radiation (PAR) for metabolism modeling from the National Ecological Observatory Network (NEON) flux tower located at the University of Kansas Field Station, about 22 km away from the farthest study site. NEON PAR data was obtained at a 1-min temporal resolution.

Sensor data from sites S1, S2, and S3 was inspected using the data cleaning tool developed by the StreamPULSE Network (<https://data.streampulse.org>, Appling et al., 2018). Analysis was performed on data collected from February 1 through 1 May 2018, as upstream releases from Tuttle Creek (Manhattan, KS; 8–14 May 2018) and Milford Lake (Junction City, KS; May 18–2 June 2018) Reservoirs contributed a large pulse of nutrients to the study reach. Non-representative readings due to sensor burial or malfunction were flagged and removed from the data set. In all, less than 15% of total data points (from 1 February through 1 May) were excluded. As some analyses (metabolism modeling, NO_3^- uptake modeling) required a complete time series, gaps were filled via linear interpolation using the StreamPULSE package for R prior to modeling (version 0.0.0.9007, Vlah & Berdanier, 2019). After modeling, values based on interpolated data were excluded from figures and statistical analysis.

Additionally, we collected samples along the study reach during sensor deployments (on 1 February, 15 February, 1 March, 15 March, 27 March, 17 April, and 2 May 2018) for ammonium (NH_4^+) and dissolved organic carbon (DOC) analysis. In the field, unfiltered water samples were collected in pre-washed 1 L Nalgene bottles and stored on ice until transported to the lab (<4 hr). In the lab, samples were filtered through $0.45 \mu\text{m}$ glass fiber filters and stored at 4°C or frozen to await NH_4^+ and DOC analysis. NH_4^+ analysis was performed following EPA method 350.1 using a SmartChem200 analyzer (Westco instruments). DOC was measured as NPOC after acidification to $\text{pH} < 2$ and 5 min of air-purging on a Shimadzu TOC-L analyzer at Kansas State University.

2.3. Metabolism Estimation

We modeled gross primary production (GPP, $\text{g O}_2 \text{ m}^{-2} \text{ d}^{-1}$) and ecosystem respiration (ER, $\text{g O}_2 \text{ m}^{-2} \text{ d}^{-1}$) at two sites, S2 and S3, using the two-station (Hall et al., 2016) and one-station (Odum, 1956) methods, respectively. We were unable to estimate metabolism at sites S0 and S1 due to considerable mixing caused by the Bowersock Dam. A one-station approach was applied at S3 because the site was sufficiently downstream (33 km) from the dam. In both modeling approaches, we used 15-min DO concentration, DO percent saturation, discharge, water temperature, and PAR data to estimate GPP and ER.

Depth (z , m) was estimated from discharge and specific channel attributes using Manning's Equation, assuming that flow is uniform and the channel is approximately rectangular:

$$Q = \frac{1}{n} \times z^{\frac{5}{3}} \times \sqrt{S_o} \times B \quad (1)$$

where, Q is discharge ($\text{m}^3 \text{ s}^{-1}$), n is the unitless Manning's roughness coefficient, S_o is the bed slope (m m^{-1}), and B is the flow width (m). Using cross-sectional profiles of the Kansas River (US Army Corps of Engineers, 1984), we determined S_o was 0.00040 m m^{-1} . We used a Manning's n of 0.040, which is typical of rivers with bends and vegetation, like the Kansas River.

Daily GPP and ER at S2 were estimated via Bayesian modeling following the two-station approach developed by Hall et al. (2016), using site S1 as the upstream location and S2 as the downstream location. We used minimally informative prior probability distributions, where $GPP \sim N(\mu = 5, SD = 10)$ and $ER \sim N(\mu = -5, SD = 10)$; Hall et al., 2016). Gas exchange (K_{600}, d^{-1}) was estimated using 24-hr Argon and N_2 dissolved gas samples collected within the release reach (as described in Burgin et al., 2018; Reisinger et al., 2016). Estimated K_{600} was then used as an informative prior for modeling (where $K_{600} \sim N(\mu = 3.40, sd = 0.13)$, Table S1 in Supporting Information S1). Mean channel depth and travel time between S1 and S2 were also used as informative priors, where travel time was estimated as the distance between S1 and S2 (5.13 km) divided by water velocity. Informative priors used in two-station model estimation are provided in Table S2 of the Supporting Information S1. Mean ER estimates were positive on 11 dates (26–28 February, 1–4 March, 10–11 March, and 14–15 March), and modeling results from these dates were excluded from analysis.

Due to the longer distance between sites S2 and S3, we modeled GPP, ER, and K at site S3 with a single-station hierarchical Bayesian model using the streamMetabolizer package for R (version 0.10.8, Appling et al., 2017). Daily gas exchange rates were pooled as a function of discharge. We assigned prior probability values of $GPP \sim N(\mu = 3.1, SD = 6)$, $ER \sim N(\mu = -7.1, SD = 7.1)$, and temperature corrected K ($K_{600}) \sim \text{Lognormal}(\mu = \text{Log}(5), \text{scale} = 0.75)$. A total of nine dates returned positive mean ER estimates (25–28 February, 1–3 March, and 10–11 March), and modeling results from these dates were excluded from analysis. We verified the appropriateness of the single-station modeling method for S3 by calculating the areal extent integrated by metabolism measurements, and comparing to the distance between the Dam and S3 (33.5 km). We estimated the distance integrated by metabolism measurements as three times the transport distance of oxygen (i.e., $3v/K$, where v is flow velocity in ms^{-1} ; Chapra & Di Toro, 1991). Flow velocity was estimated as Q divided by river cross-sectional area (A in m^2 ; $v = Q/A$). We defined A as the product of z and approximate river width (w) estimated from USGS/NASA Landsat aerial images ($A = z \times w$). The mean areal extent of metabolism measurements at S3 was 28.8 km (where $v = 0.46 ms^{-1}$, $K = 4.29 d^{-1}$).

Predicted areal N uptake rates were calculated from metabolism estimates and stoichiometric ratios (Hall & Tank, 2003) as

$$U_{\text{Auto}} = \frac{GPP \times 0.5}{20} \quad (2)$$

$$HR = |ER| - 0.5 \times GPP \quad (3)$$

$$U_{\text{Het}} = \frac{0.2 \times HR}{1 - 0.2} \times \frac{1}{20} \quad (4)$$

$$U_{\text{Pred}} = U_{\text{Auto}} + U_{\text{Het}} \quad (5)$$

where, U_{Auto} , U_{Het} , and U_{Pred} are the autotrophic, heterotrophic, and total N uptake rates ($g N m^{-2} d^{-1}$), respectively, and HR is heterotrophic respiration ($g O_2 m^{-2} d^{-1}$). We assumed U_{Auto} and U_{Het} were driven by carbon (C) production (i.e., a photosynthetic quotient of 1; Hall & Tank, 2003), and assumed net autotrophic production was half of GPP (as assumed by Odum, 1957; Webster & Meyer, 1997). We also assumed a molar C:N ratio of 20, based on an average of epilithon and filamentous algae C:N ratios from open-canopy streams (Hall & Tank, 2003). To estimate HR , we multiplied GPP by a heterotrophic growth efficiency coefficient of 0.2, which corresponds to a moderate growth efficiency (Hall & Tank, 2003), and subtracted this from ER. From HR we estimated U_{Het} by assuming a photosynthetic quotient of 1 and a molar C:N ratio of 20, following the assumptions made by Hall and Tank (2003). Lastly, U_{Pred} was computed as the sum of N utilized in autotrophic assimilation and heterotrophic respiration.

2.4. Nitrate Uptake Calculations

Assimilatory NO_3^- uptake (U_{Diel} , $g-N m^{-2} d^{-1}$) was calculated following the extrapolated diel method (Heffernan & Cohen, 2010) as

$$U_{\text{Diel}} = \frac{Q}{SA} \sum_{t=0}^{24} \left(\left[\text{NO}_3^- \right]_{\text{max}(t=0)} - \left[\text{NO}_3^- \right]_t \right) \quad (6)$$

where SA is riverbed surface area (m^2), t is time of day starting at dawn (hr), $[\text{NO}_3^-]_{\text{max}(t=0)}$ is pre-dawn maximum NO_3^- concentration (mg N L^{-1}), and $[\text{NO}_3^-]_t$ is concentration at time t (mg N L^{-1}). A two-hour smoothing window was used to capture dominant temporal trends in the 15-min nitrate data and reduce the influence of transient sensor fluctuations. Daily U_{Diel} was calculated as the difference between the pre-dawn maximum NO_3^- concentration ($[\text{NO}_3^-]_{\text{max}(t=0)}$, mg-N L^{-1}) and NO_3^- concentration at time t ($[\text{NO}_3^-]_t$, mg-N L^{-1}), summed from the time of the maximum pre-dawn concentration ($t = 0$) to the time of the following day's pre-dawn maximum ($t = 24$). This value was then scaled to the reach by multiplying by mean daily river discharge (Q , $\text{m}^3 \text{s}^{-1}$), dividing by riverbed surface area (SA , m^2), and applying a conversion factor of 86,400 s per day. At each site, riverbed surface area was equal to the width of the riverbed channel (at S1, S2, or S3) multiplied by the distance between the waste release point and the end of the study reach (31 km). Lastly, we transformed U_{Diel} to uptake velocity (V_f , mm min^{-1}) by dividing U_{Diel} by mean daily NO_3^- :

$$V_f = \frac{U_{\text{Diel}}}{[\text{NO}_3^-]} \quad (7)$$

2.5. Hydrologic Modeling

To parameterize the transport of the NO_3^- plume, from the waste release through the study reach, we modeled dispersion using the advection-dispersion equation (ADE) under mean flow conditions (Van Genuchten et al., 2013). We assumed: (a) flow was uniform in the longitudinal direction, (b) water velocity and nutrient concentration were constant with depth, (c) lateral dispersion was much greater than longitudinal dispersion, and (d) NO_3^- transport was conservative. We recognize that NO_3^- transformations are occurring in the Kansas River, but assuming conservative transport in this model allowed us to obtain the most conservative estimate of mixing length distance. The model was evaluated for average conditions under the steady-state assumption (Text S1, Table S2 in Supporting Information S1). Under mean flow conditions, the nutrient plume was fully mixed 24 km downstream of the addition site but was approximately fully mixed (>80%) by 12.50 km downstream (Figure S1 in Supporting Information S1).

2.6. Statistical Testing

We used a two-sided, paired sample Student's T-test to evaluate the difference between means of GPP at S2 and GPP at S3 during dates where metabolism data at S2 was available. The same approach was used to test whether ER was different between S2 and S3. One-way analysis of variance (ANOVA) was used to evaluate differences between U_{Diel} or V_f at S1, S2, and S3. Pearson correlation using complete observations of parameters was used to evaluate the correlation between U_{Diel} or V_f at each site and day of year, U_{Diel} and U_{Pred} at each site, and U_{Diel} or V_f and GPP or ER. Statistical testing was done using R software (version 3.6.1; R Project for Statistical Computing, Vienna, Austria) using the default statistical testing package *stats*.

3. Results

The highest NO_3^- concentrations were measured nearest the waste release point (S1, Figure 2). From 26 February 2018 onwards, upstream (S0) NO_3^- concentrations were below sensor detection limits ($<0.1 \text{ mg-N L}^{-1}$, verified by grab sampling; Figure 2a). NO_3^- concentrations at S1 averaged 5.52 mg-N L^{-1} during the release, and 2.82 mg-N L^{-1} after the release was halted on 1 April 2018 (Figure 2a). S3 experienced the second highest concentrations, with 0.63 mg-N L^{-1} during release and 0.22 mg-N L^{-1} post release. S2 experienced the lowest concentrations downstream of the release point (mean = 0.33 mg-N L^{-1} during, 0.23 mg-N L^{-1} post; Figure 2a). Patterns of nitrate load resembled patterns of nitrate concentration (Figure S3 in Supporting Information S1). Diel fluctuations in 15-min NO_3^- concentrations were evident at each site during the study period (Figures 2c–2e). Dissolved organic carbon (DOC) concentrations were highest at S1 (mean = 11.2 mg L^{-1}), which was on average almost twice as DOC-rich as upstream (S0 mean = 5.88 mg

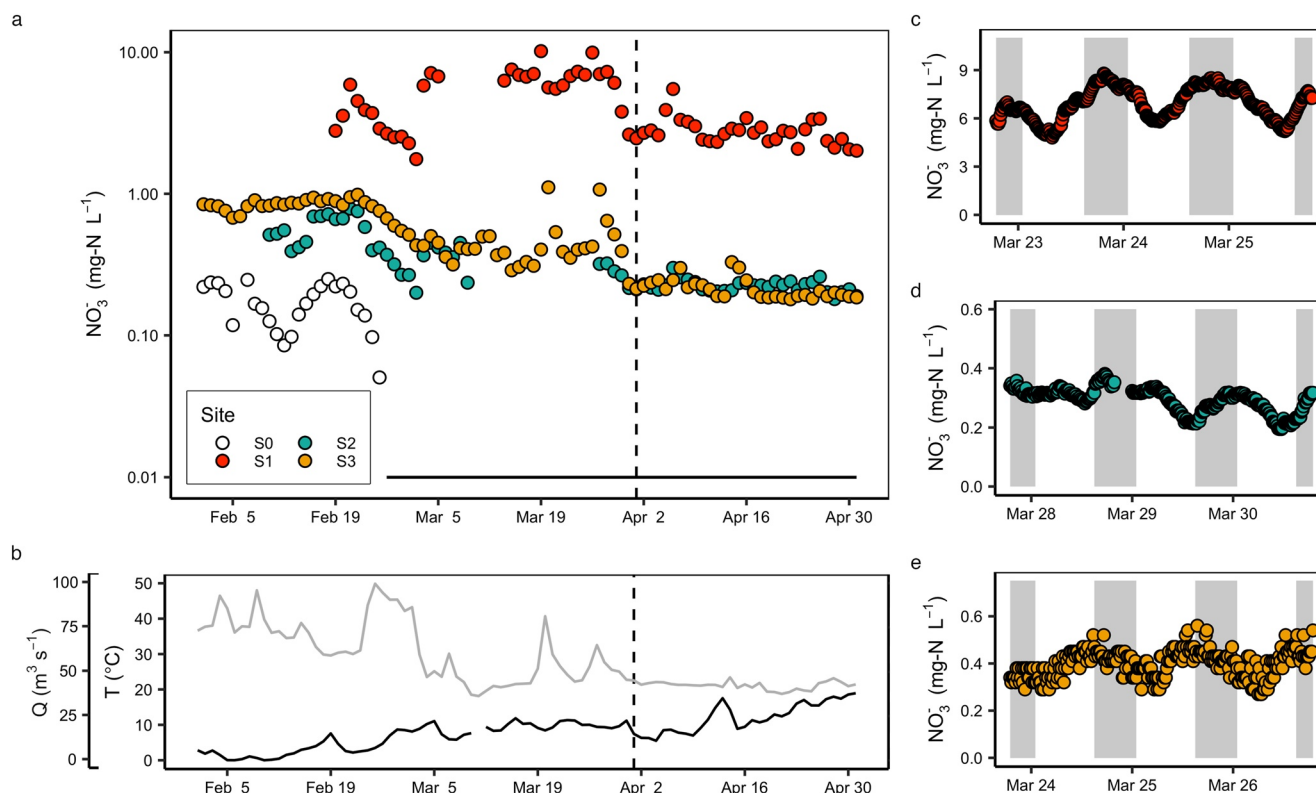


Figure 2. Mean daily nitrate (NO_3^-) concentration at each study site (a), mean daily water temperature (T, black line) and discharge (Q, gray line) at site S3 (b), and select dates of 15-min NO_3^- measurements at site S1 (c), S2 (d), and S3 (e). Vertical dashed line (a, b) denotes end date of release (1 April 2018). Colored points in (a, c, d, e) denote study site, with white signifying S0, red = S1, blue = S2, and yellow = S3. Solid black line from 26-February to 30-April in (a) denotes period in which NO_3^- was below detection limit ($<0.1 \text{ mg-N L}^{-1}$) at site S0. Gray bars in (c–e) denote local night-time (8:00 p.m. - 6:00 a.m.). Note that the y-axis of (a) is displayed on a log scale.

L^{-1} , Table 1). Water temperature ranged from a minimum of 0°C to a maximum of 18.9°C , with mean of 8.2°C (Figure 2b). Discharge was variable over the course of the study (S3 mean = $56.0 \text{ m}^3 \text{s}^{-1}$, standard deviation = $17.4 \text{ m}^3 \text{s}^{-1}$; Table 1, Figure 2b), but was historically low for the region (annual mean from 1917 to 2018 = $194 \text{ m}^3 \text{s}^{-1}$, USGS De Soto gauge data) due to a drought. A temporary discharge increase and NO_3^- decrease occurred in late February and was the result of a controlled reservoir release from Perry Lake by the US Army Corps of Engineers (Figure 2b.)

GPP and IER near the release point (S2) and downstream (S3) were similar (Figure 3). GPP at S2 averaged $5.72 \text{ g O}_2 \text{ m}^{-2} \text{d}^{-1}$, while GPP at S3 during the same time period averaged $5.71 \text{ g O}_2 \text{ m}^{-2} \text{d}^{-1}$ (Table 2). During the same time period, IER was slightly elevated further downstream ($3.18 \text{ g O}_2 \text{ m}^{-2} \text{d}^{-1}$ at S2, $3.89 \text{ g O}_2 \text{ m}^{-2} \text{d}^{-1}$ at S3, Table 2). This resulted in a GPP to ER ratio (P:R) that was nearly twice as high at S2 as it was at S3 (3.28 vs.

Table 1
Mean Parameters at Each Study Site From 1 February to 1 May 2018, With Standard Deviation in Parentheses

Site	Q ($\text{m}^3 \text{s}^{-1}$)	NH_4^{+} (mg-N L^{-1})	NO_3^- (mg-N L^{-1})	DOC ^a (mg L^{-1})	U_{Diel} (g N $\text{m}^{-2} \text{d}^{-1}$)	V_f (mm min^{-1})
S0	50.4 (17.5)	0.05 (0.03)	0.05 ^b (0.08 ^b)	5.88 (0.49)	NA	NA
S1	50.5 (17.6)	0.05 (0.02)	4.17 (2.09)	11.2 (12.4)	4.84 (3.85)	0.98 (0.80)
S2	50.5 (17.6)	0.06 (0.03)	0.32 (0.18)	5.78 (0.52)	0.49 (0.36)	1.11 (0.63)
S3	56.0 (17.4)	0.08 (0.06)	0.49 (0.27)	6.24 (0.84)	0.64 (0.58)	0.85 (0.73)

Note. Q, discharge, NH_4^{+} , ammonium, NO_3^- , nitrate, DOC, dissolved organic carbon, U, nitrate uptake, V_f , nitrate uptake velocity, NA, data not collected.

^aValue is an average of periodic grab sample measurements. ^bValue is below sensor detection limit of 0.1 mg-N L^{-1} .

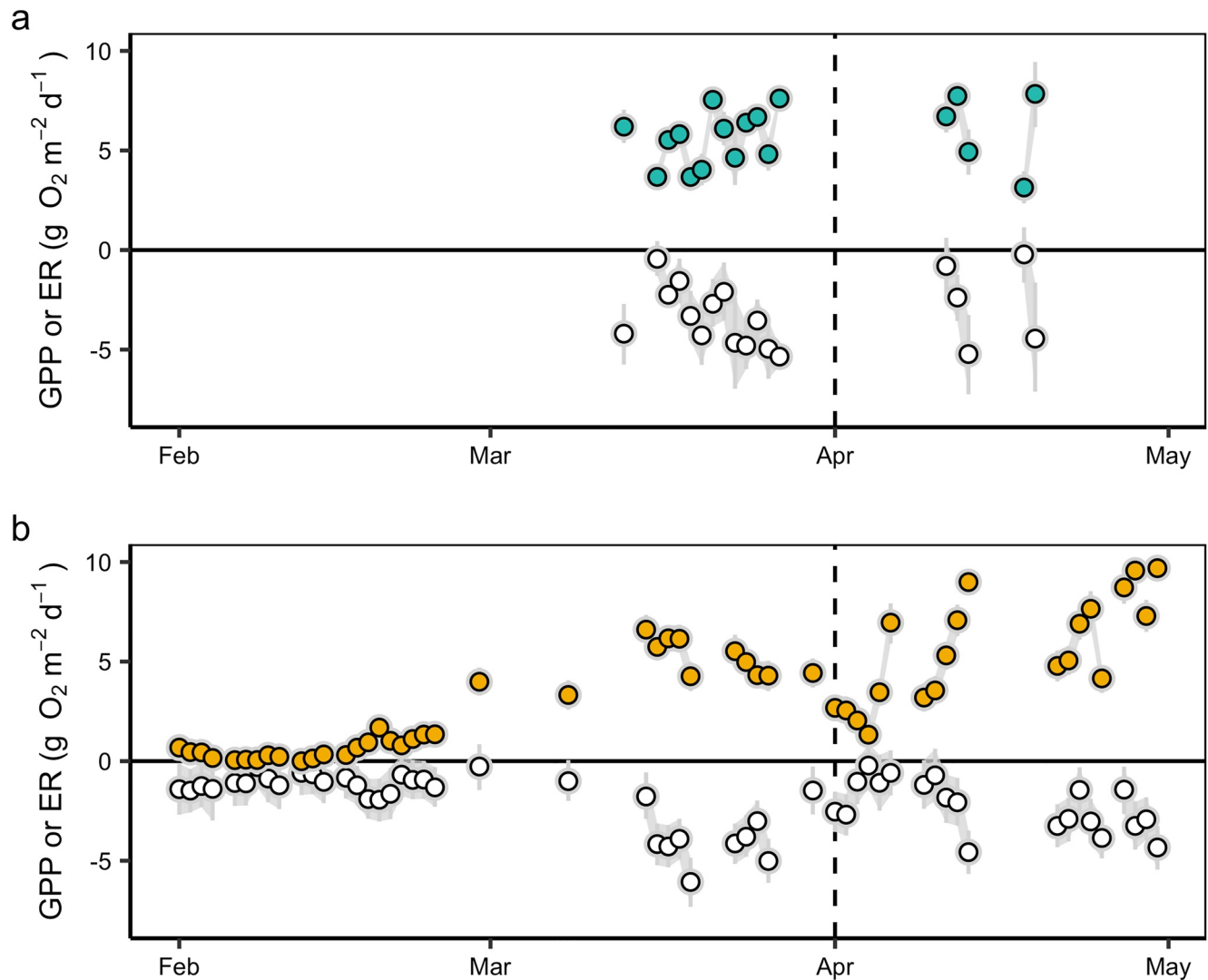


Figure 3. Modeled estimates of gross primary productivity (GPP) and ecosystem respiration (ER) at S2 (a) and S3 (b). Gray bars denote the 95% credible interval of model estimates, colored points denote the median model value for one day.

1.67, Table 2). At both sites, GPP and $|ER|$ were lowest during February, increased during March and mid-April, and peaked in late April (Figure 3b).

NO_3^- uptake (U_{Diel}) was greatest nearest the release point (Figure 4). U_{Diel} was $\sim 10\text{--}100\times$ higher at the release (site S1, $F = 21.53$, $p < 0.001$; Figure 4a) than at downstream sites (S2, S3). Mean U_{Diel} at S1 was $4.84 \text{ g N m}^{-2} \text{ d}^{-1}$, at S2 was $0.49 \text{ g N m}^{-2} \text{ d}^{-1}$, and at S3 was $0.64 \text{ g N m}^{-2} \text{ d}^{-1}$ (Table 1). U_{Diel} decreased from February to May at S2 ($R^2 = 0.22$, $p < 0.001$) and S3 ($R^2 = 0.42$, $p < 0.001$), but was constant through time at

Table 2

Mean Metabolism Estimates for Each Site From 1 February to 1 May 2018, With Standard Deviation in Parentheses

Site	Method	GPP ($\text{g O}_2 \text{ m}^{-2} \text{ d}^{-1}$)	$ ER $ ($\text{g O}_2 \text{ m}^{-2} \text{ d}^{-1}$)	P/R	K (d^{-1})
S2	Two-Station Bayesian	5.72 (1.51)	3.18 (1.69)	3.28 (3.56)	3.40 (0.002)
S3	Single-Station Bayesian	3.45 ^a (2.91)	2.03 ^a (1.41)	2.10 ^a (2.70)	4.29 (0.97)

Note. GPP, gross primary productivity, $|ER|$, absolute value of ecosystem respiration, P/R, ratio of GPP to $|ER|$, K, Reaeration rate derived from metabolism models.

^aMean values during the duration of S2 data: GPP = $5.71 \text{ g O}_2 \text{ m}^{-2} \text{ d}^{-1}$, $|ER|$ = $3.89 \text{ g O}_2 \text{ m}^{-2} \text{ d}^{-1}$, P/R = 1.67.

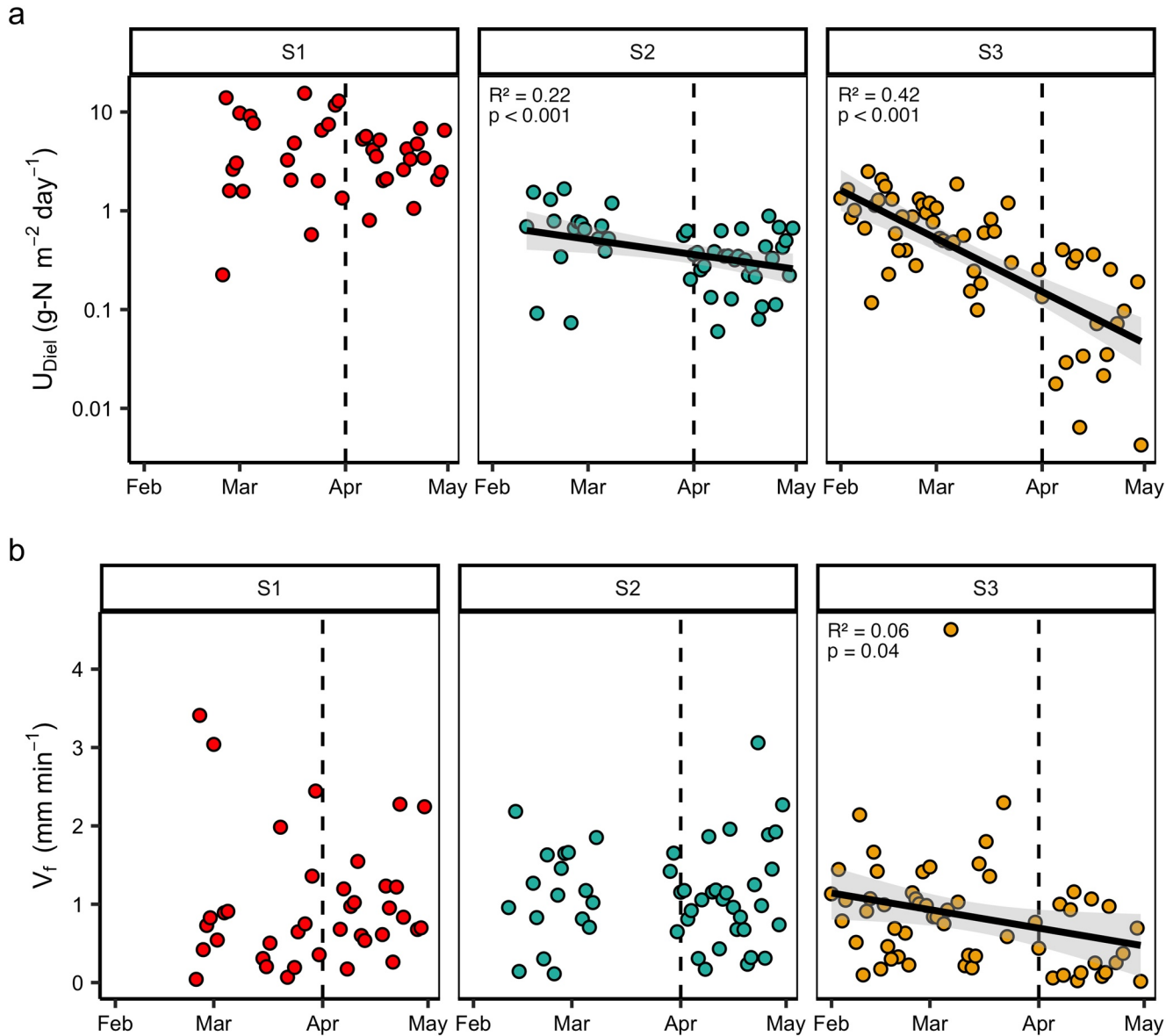


Figure 4. Nitrate uptake calculated from diel measurements (U_{Diel}) for S1, S2, and S3 (a) and nitrate uptake velocity (V_f) for S1, S2, and S3 (b). Colored points denote data for one day. Vertical dashed line denotes end date of release (1 April 2018). Linear trend line displayed when significant (p -value of linear regression < 0.05), with gray shading signifying 95% confidence interval of the regression estimate. Note that the y -axis of (a) is displayed on a log scale.

S1 ($R^2 = 0.007$, $p = 0.27$). V_f was not significantly different between sites ($F = 1.83$, $p = 0.17$; Figure 4b). V_f decreased through time at S3 ($R^2 = 0.06$, $p = 0.04$), but was constant through time at S1 ($R^2 = -0.03$, $p = 0.96$) and S2 ($R^2 = -0.02$, $p = 0.65$).

Predicted NO_3^- uptake from stoichiometry (U_{Pred}) more closely matched sensor-measured uptake (U_{Diel}) after the waste release concluded at both sites S2 and S3 (colored symbols on Figure 5). In contrast, during the waste release, U_{Diel} was generally much higher than uptake predicted by stoichiometry (U_{Pred}) at S3 (open symbols, Figure 5). U_{Diel} was negatively related to predicted uptake rates from stoichiometry (U_{Pred}) at site S3 ($R^2 = 0.34$, $p < 0.01$, Figure 5b), but unrelated at S2 ($R^2 = -0.12$, $p = 0.50$, Figure 5a). At S3, during the waste release U_{Diel} was on average 13 times greater than U_{Pred} (mean = $0.85 \text{ g N m}^{-2} \text{ d}^{-1}$ vs. $0.06 \text{ g N m}^{-2} \text{ d}^{-1}$). However, there was no significant correlation between U_{Diel} and U_{Pred} during the release ($R^2 = 0.11$, $p = 0.08$). After the release ended, U_{Diel} became nearly equivalent to U_{Pred} (mean = $0.14 \text{ g N m}^{-2} \text{ d}^{-1}$ vs. $0.13 \text{ g N m}^{-2} \text{ d}^{-1}$); and the relationship between U_{Diel} and U_{Pred} was not statistically significant ($R^2 = 0.11$, $p = 0.17$).

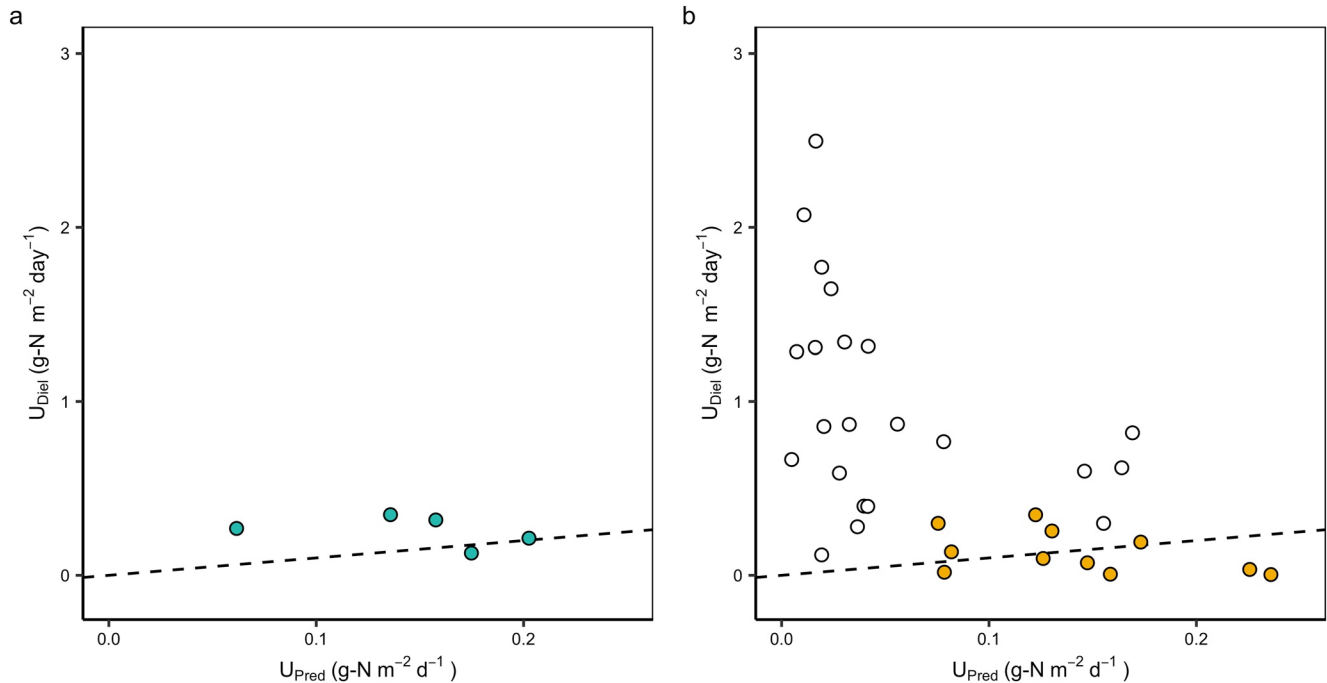


Figure 5. Relationship between predicted N uptake from stoichiometrically scaled metabolism estimates (U_{Pred}) and nitrate uptake calculated from diel measurements (U_{Diel}) at site S2 (a) and S3 (b). Dotted line is a 1:1 reference line. Open circles denote data during the waste release period (February–March); colored points denote data after the waste release period (April).

NO_3^- uptake (U_{Diel}) was not coupled to metabolism (GPP, ER) at either site during or after the waste release (Figure 6). GPP was negatively related to U_{Diel} at S3 ($R^2 = 0.34$, $p < 0.01$; Figure 6b), and GPP and U_{Diel} were unrelated at S2 ($R^2 = -0.02$, $p = 0.41$; Figure 6a). Similarly, IERI was negatively related to U_{Diel} at S3 ($R^2 = 0.19$, $p < 0.01$; Figure 6d) and unrelated at S2 ($R^2 = -0.32$, $p = 0.86$, Figure 6c). V_f was not related to metabolism at any site. GPP was not related to V_f at S2 ($R^2 = -0.11$, $p = 0.49$, Figure S2a in Supporting Information S1) or S3 ($R^2 = 0.02$, $p = 0.19$, Figure S2b in Supporting Information S1). IERI was not related to V_f at S2 ($R^2 = -0.32$, $p = 0.89$; Figure S2d in Supporting Information S1), or S3 ($R^2 = -0.004$, $p = 0.36$; Figure S2c in Supporting Information S1).

4. Discussion

We asked how the relationship between nutrient uptake and metabolism would change over time and space in response to a unique nutrient addition in a large river. We hypothesized that (a) nitrate uptake would be lowest near the release point due to nutrient saturation, (b) gross primary production would be decoupled from uptake at the release point due to nutrient saturation, and (c) the positive relationship between nutrient uptake and primary productivity would be strengthened after the conclusion of the nutrient addition, as biota were no longer nutrient saturated. Contrary to hypothesis 1, elevated N concentrations from the waste input (Figure 2a) lead to the greatest levels of nutrient uptake near the source (Figure 4a). In fact, consistently high nutrient uptake occurred during the release period, despite low water temperatures and depressed metabolic activity. Nutrient uptake was not coupled to gross primary production (Figures 6a and 6b) or ecosystem respiration (Figures 6c and 6d) at any time, contrary to hypothesis 2 and 3. Instead, GPP and IERI were either unrelated or negatively related to uptake (Figure 6). In this discussion, we contextualize our observations of NO_3^- uptake and stream metabolism and explore the possible drivers of observed decoupling.

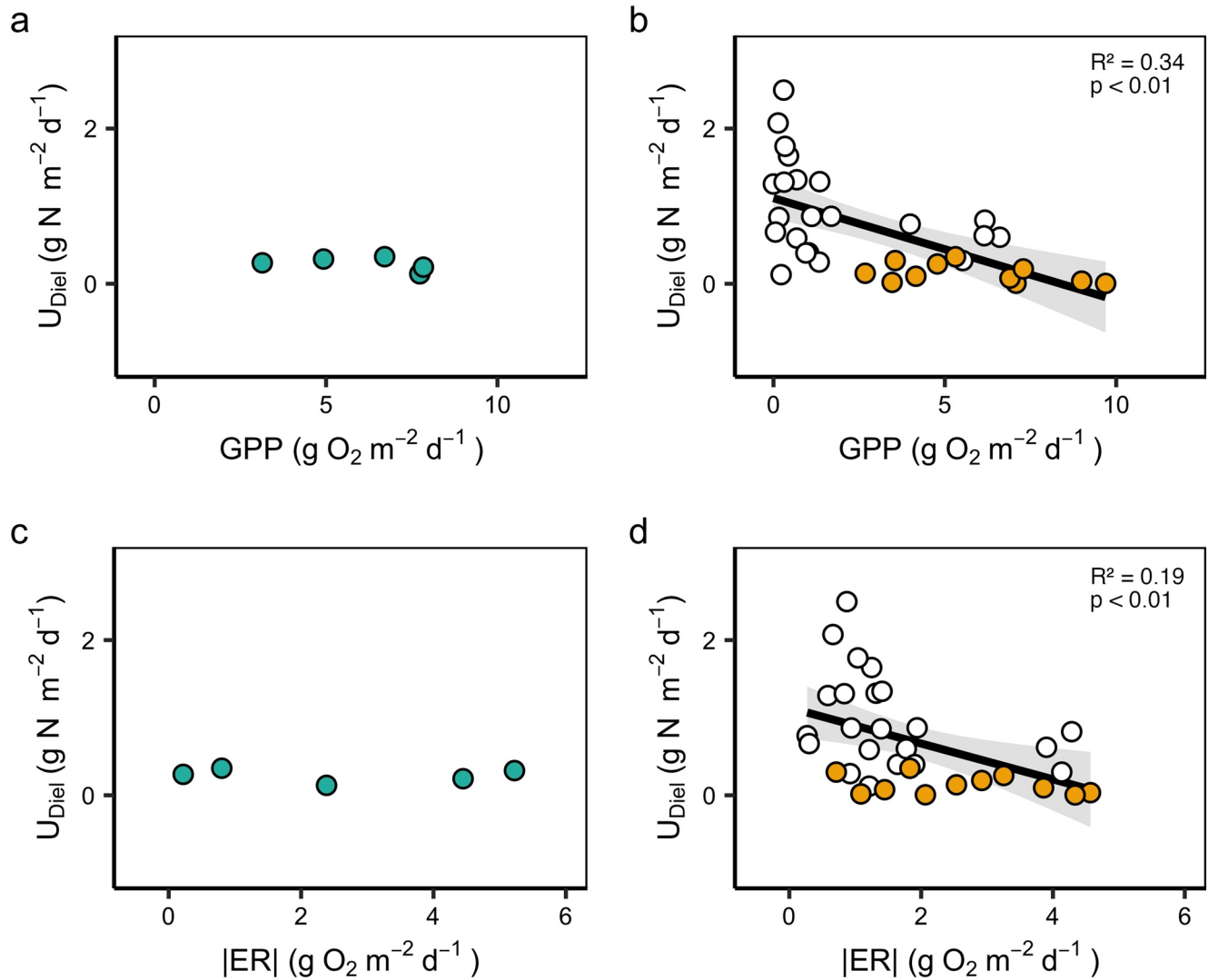


Figure 6. Relationships between gross primary productivity (GPP) and nitrate uptake calculated from diel measurements (U_{Diel}) at site S2 (a) and S3 (b) and between ecosystem respiration (ER) and U_{Diel} at site S2 (c) and S3 (d). Open circles denote data during the waste release period (February–March); colored points denote data after the waste release period (April). Linear regression line added for statistically significant relationships ($p < 0.05$); 95% confidence interval of regression denoted by gray shaded area. Regression statistic (R^2 , p) corresponds to all data points at the site.

4.1. Nitrate Uptake

Knowledge of total NO_3^- uptake is still constrained by the lack of studies conducted in larger ($Q > 1 \text{ m}^3 \text{ s}^{-1}$) rivers (Tank et al., 2008). We synthesized meta-analysis data (Tank et al., 2008) to provide context for our large river results to other, smaller systems (Figures 7b and 7d). Generally, NO_3^- uptake rate (U) and uptake velocity (V_f) increase with higher river discharge, but the vast majority of uptake studies are conducted in lower discharge ($< 1 \text{ m}^3 \text{ s}^{-1}$) systems. Despite this, our uptake rates were well predicted by the linear regression presented in Tank et al. (2008), and V_f in this study was similar to that of other large rivers (Rode et al., 2016; Hensley et al., 2014; Figure 7d).

Our results indicated that higher nutrient concentrations promoted higher uptake rates, as also observed by Hall et al. (2009). Unlike streams in other studies (e.g., Covino et al., 2010; Earl et al., 2006), which show uptake reaching saturation at high NO_3^- concentrations, we did not observe uptake to reach a plateau as concentrations increased. We expected biota to be saturated under high N supply, causing the lowest uptake velocities nearest the release point (Dodds et al., 2002). Instead, higher uptake velocities nearer the release point suggest that the

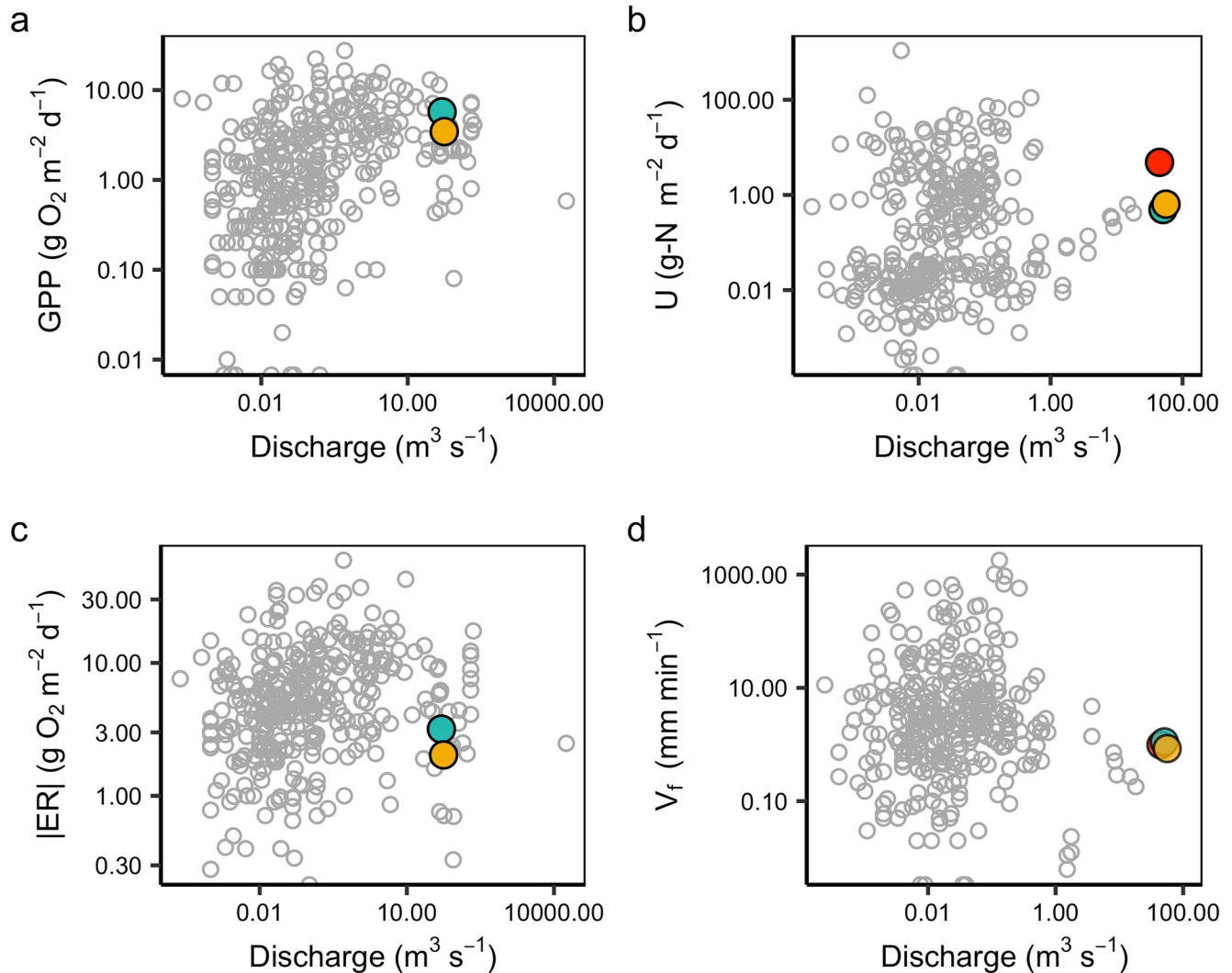


Figure 7. Meta-analysis of previously published results of GPP (a), ER (b), U (c), and V_f (d) compared to mean results from this study (colored points, where U from this study is U_{Diel}); gray points are meta-analysis data summarized by Hall et al. (2016); (a and b), Tank et al. (2008), where $U = V_f/[NO_3^-]$; (b and d), plus data from Rode et al. (2016) and Hensley et al. (2014). Red points signify data from site S1 of this study, blue = site S2, yellow = site S3. Note that the x and y-axis of (a–d) are displayed on a log scale.

waste release accelerated nutrient processing in the Kansas River. Our present data set identifies this trend, but cannot resolve the specific drivers of increased uptake, thus further analysis will be necessary to understand the effect and duration of these drivers.

Three possible, non-mutually exclusive explanations exist for this unexpected response in V_f : (a) a lag between the predicted biological response and the start of stream monitoring, (b) biota in proximity to the release source may already be acclimated to high N supply, and/or (c) alternative uncertainties and limitations in the approach of estimating ecosystem-level fluxes. To the first point, we were unable to install equipment and begin sampling until ~3 months into the waste release. The City required a period of time to work out the waste disposal method in a manner that made pumping approximately continuous. Thus, we may have missed an adjustment period when biota were initially saturated, and captured what was increased uptake over time after acclimating to high N supply (as the environment transitioned from low to high N, as in Ribot et al., 2013). Additionally, V_f remained elevated one month following the release at the two sites nearest the release point and decreased at the site furthest downstream (Figure 4b), suggesting that the release may have had a more lasting impact on the capacity for nutrient uptake at S1 and S2, and indicating a lag in biological response to cessation of the nutrient addition.

Alternatively, to the second point, in-stream biota may have already been acclimated to higher N supply. Despite the clear increase in nitrate concentration due to the waste release (median NO_3^- at S3 the month preceding the release = 0.18 mg-N L^{-1}), NO_3^- levels were not out of historic range (at USGS De Soto gage from start of record on 1 October 1990 through 1 January 2017, median NO_3^- = 1.02 mg-N L^{-1} , maximum = 4.74 mg-N L^{-1}). To the third point, scaling up any point-based sensor measurement to an ecosystem-level flux will create unavoidable error, due to streambed heterogeneity (Reijo et al., 2018), variation in activity within the reach itself (Reichert et al., 2009), or even the influence of riparian activity (as observed by Dodds et al., 2017).

4.2. Stream Metabolism

Understanding large river metabolism has become a focus of the field in recent years, as synthesis work has allowed for better understanding of the relationship between stream size and system energetics (Dodds et al., 2013; Hall et al., 2016; Young et al., 2008). The Kansas River, an eighth order river, adds a valuable large river data point to syntheses of whole river metabolism. We found the Kansas River to be autotrophic at sites S2 and S3 (Table 2). For context, Dodds et al. (2008) measured whole-stream metabolism in the Kansas River and found the system to be dominantly heterotrophic ($\text{GPP} = 8.48 \text{ g O}_2 \text{ m}^{-2} \text{ d}^{-1}$, $|\text{ER}| = 12.12 \text{ g O}_2 \text{ m}^{-2} \text{ d}^{-1}$); this discrepancy may be due in part to the seasonal or site differences between our two studies (the Dodds et al. study was conducted 180 km upstream of our sites and during autumn). GPP was relatively high in this study in comparison to other rivers (Figure 7a), but was well predicted by the general relationship between stream size and metabolism as explored by Hall et al. (2016). This suggests that the river was highly productive, despite cold water temperatures in February (mean = 2.9°C) and March (mean = 9.2°C). As cold should have suppressed primary productivity's response to increased nutrient loading, we can infer that the river may have been exceptionally productive if water temperatures were warmer. Additionally, hydrology was considerably less stable during February and March ($\text{SD} = 17 \text{ m}^3 \text{ s}^{-1}$; Figure 2b) than during April ($\text{SD} = 2.3 \text{ m}^3 \text{ s}^{-1}$), which may have inhibited the growth of more established primary producer communities.

Ecosystem respiration was within the low to medium range of values summarized by Hall et al. (2016), but similar to ER measured in other higher order rivers (Figure 7c). ER rates are greatly affected by benthos substrate type, available organic matter, and hyporheic zone connectivity (Young et al., 2008). The Kansas River is a low-land river with a highly homogeneous, sandy substrate; thus, hyporheic zone flow and sediment redox conditions are unlikely to be significantly different between sites. This suggests that the high to low ER gradient observed between sites is not primarily driven by either of these two factors. Dissolved organic carbon (DOC, Table 1) from the release may have driven higher heterotrophic respiration rates, causing ER to peak nearest the release point and decrease downstream. The waste release did contain a significant DOC input: While not as dramatic a difference as the nitrate and ammonium concentrations, DOC concentration was twice as high in the pond water than the river, and this could have supported an increase in assimilatory N uptake by heterotrophs (as in Bernhardt & Likens, 2002).

4.3. Metabolism and NO_3^- Dynamics

Stoichiometric estimates of areal uptake (U_{Pred}) did not estimate uptake measured by diel variation using sensors (U_{Diel} , Figure 5) as we predicted. At S3, during the waste release (Figure 5b white points), U_{Diel} was higher than predicted NO_3^- uptake from metabolism and growth assumptions. After the release period ended (Figure 5b yellow points), U_{Pred} became a much more accurate predictor of U_{Diel} . This changing relationship through time created an overall negative correlation between U_{Pred} and U_{Diel} . The mismatch between predicted uptake rates and measured uptake rates during the waste release at S3, and throughout the study at S2, suggests that processes other than assimilation may be responsible for the higher measured uptake. However, our estimates of whole-river denitrification indicate that these rates were low ($<1 \text{ mg-N m}^{-2} \text{ hr}^{-1}$, Burgin et al., 2018), suggesting that denitrification was not a large sink of NO_3^- . Predicted uptake was calculated using assumed production and respiration efficiencies, and an assumed molar C:N ratio of the epilithion and filamentous algal communities, which may not be representative of communities in the Kansas River. However, as predicted uptake would need to increase by more than 70% at S2 and 85% at S3 to match measured uptake, this disparity is unlikely to be due to calculation assumptions alone. Another possible explanation for the discrepancy between U_{Pred} and U_{Diel} is that the algal and microbial communities responsible for NO_3^- uptake during the waste release may not abide by the stoichiometric growth assumptions used here. Additionally, the presence of other forms of nitrogen (i.e., NH_4^+) may have

impacted estimates of U_{Pred} . If NH_4^+ was available, it would likely be preferentially assimilated into auto- and heterotrophic biomass before NO_3^- . In this case, preferential NH_4^+ assimilation would contribute to a decrease in C:N, causing a discrepancy between U_{Pred} and U_{Diel} .

We found negative relationships between U_{Diel} and GPP and U_{Diel} and ER at S3, furthest from the discharge point (Figures 6b and 6d). Similar to the relationship between U_{Diel} and U_{Pred} , high U_{Diel} during the waste release (Figure 6 white points) and comparatively lower GPP and |ER| during the same period suggests that processes other than autotrophic or heterotrophic assimilation were responsible for high NO_3^- removal during this time. Nearer the release at site S2, metabolism was not correlated to U_{Diel} , but the limited overlap in dates of two-station metabolism and U_{Diel} data ($n = 5$) at this site limits our ability to draw a robust conclusion. Other studies have shown a positive relationship between NO_3^- uptake and GPP (T. P. Covino et al., 2018; Heffernan & Cohen, 2010; Lupon et al., 2016), but these studies have been conducted under relatively low ($<1 \text{ mg NO}_3^- \text{-N L}^{-1}$) nutrient conditions. To our knowledge, only one other study has investigated the relationship between NO_3^- uptake and primary productivity in an N-rich (mean $3.66 \text{ mg NO}_3^- \text{-N L}^{-1}$) river, finding a positive relationship between the two parameters only when discharge was low ($<0.45 \text{ m}^3 \text{ s}^{-1}$, Rode et al., 2016).

An additional explanation for the decoupled relationship between uptake and metabolism is that the waste addition may have enhanced other biogeochemical transformation rates. Nitrification (NH_3 or $\text{NH}_4^+ \rightarrow \text{NO}_3^-$) may distort the diel NO_3^- signal (Harrison et al., 2005), driving an increase in NO_3^- as biota are simultaneously removing it through uptake pathways. This may confound estimates of U_{Diel} , as this calculation assumes that all diel change in NO_3^- concentration is due to uptake processes. Since ammonium concentrations at the release point were not different from upstream concentrations (Table 1), it's unlikely that the waste release significantly increased riverine nitrification. Preliminary nitrification potential assays on both pond water and river water returned low to undetectable rates, so we did not pursue further investigation of nitrification. However, especially given the low ambient ammonium concentrations, benthic ammonium uptake may have been rapid in the relatively small distance between the source ponds and the waste outfall sensor location. Additionally, if DOC from the release pulse (Table 1) increased heterotrophic respiration rates, the subsequently elevated ER may be then decoupled from estimates of NO_3^- uptake.

5. Conclusions

In this study, a controlled nitrogen waste release by the City of Lawrence, KS, dramatically increased nitrate availability in the Kansas River. We hypothesized that nutrient uptake would be saturated in response to an extreme increase in nutrient supply, but we did not find evidence to support uptake saturation. High uptake rates were observed along the study reach, with the highest rates occurring where nutrient supply was also highest, regardless of ambient N concentration. Consistently high nitrate uptake rates signified that the study reach had a high capacity for nutrient removal. Unlike general trends observed in the literature, NO_3^- uptake was high when gross primary productivity and ecosystem respiration were relatively low, suggesting that high NO_3^- removal was not due to assimilatory activity alone. To our knowledge, this is the largest river where NO_3^- uptake rates have been measured, adding to the growing body of work investigating nutrient retention in large rivers. Overall, our work further highlights the lack of knowledge, and need for additional study, regarding the mechanisms that regulate nutrient retention in large rivers.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

High-frequency sensor data from sites S1, S2, and S3 are available on the StreamPULSE database under site codes “KANSASREASTLAWRENCE” (S1), “KANSASRFALLLEAF” (S2), and “KANSASR” (S3) at <https://data.streampulse.org/>. All other data, code, and model outputs from this analysis are available on GitHub at <https://doi.org/10.5281/zenodo.5539473>.

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